Q.no	List of Questions (Practice SET_DBMS) MTE
Q.no 1	Define DBMS. List any four advantages & disadvantages of a Database Management System over a traditional File System.
Q.no 2	Define physical and logical data independence. Explain their significance in a database system with a practical scenario.
Q.no 3	Explain the primary differences between a file-processing system and a modern Database Management System (DBMS).
Q.no 4	Define 'schema' and 'instance' in a database. Illustrate the difference using one example each.
Q.no 5	Define Data Definition Language (DDL). List any five DDL commands along with their purpose.
Q.no 6	Define relation, tuple, attribute and domain as used in the relational data model.
Q.no 7	Explain the difference between external, conceptual, and internal levels of database architecture.
Q.no 8	Why must a primary key be unique and non-null? Explain both requirements with reference to entity identification.
Q.no 9	Define entity integrity, referential integrity, key constraint, and domain constraint.
Q.no 10	Explain the meaning of relationship degree in an ER model and give examples of unary, binary, and ternary relationships.
Q.no 11	Explain how entity integrity and referential integrity differ, using Student and Enrollment relations.
Q.no 12	Compare DDL, DML, and DCL. Give two representative commands for each category and explain their roles.
Q.no 13	Identify suitable SQL constraints for Student(Student_ID, Email, Age, Program) to enforce uniqueness, required values, and valid age values.
Q.no 14	A relation Employee(EID, Email, Phone, Name) has unique EID, Email, and Phone. Explain how candidate and alternate keys would be selected.
Q.no 15	Analyze a case where an Enrollment record contains a Student_ID that does not exist in Student. Which constraint is violated and why?
Q.no 16	Design integrity constraints for a Bank Management System containing Customer and Account relations. Include primary-key, foreign-key, and domain constraints.
Q.no 17	Evaluate how integrity constraints prevent data inconsistency in a University database. Give examples involving primary keys, foreign keys, and domain restrictions.
Q.no 18	Given relations STUDENT(RollNo, Name, DeptID) and DEPT(DeptID, DeptName), write SQL queries to (a) list all students belonging to the 'CSE' department, and (b) find the department-wise count of students.
Q.no 19	Formulate Relational Algebra expressions for Projection, Selection, and Cartesian Product. Explain their mathematical properties.
Q.no 20	Compare the use of relational algebra, relational calculus, and SQL for expressing a multi-table University query.
Q.no 21	Differentiate between a Candidate Key, Super Key, Primary Key, and Foreign Key. Explain referential integrity violations.
Q.no 22	Write SQL statements to (a) create a VIEW displaying employees with salary greater than 60,000, and (b) create an INDEX on the Salary column of the EMP table.
Q.no 23	The proposed database consists of the following relations: •Employee (EmpID, Name, Department, Salary, Gender) •Project (ProjID, ProjName, Budget, Location) •WorksOn (EmpID, ProjID, HoursWorked) •PastEmployee (EmpID, Name, Department, Salary, Gender) Formulate the relational algebra query: a. Find all details of the employee who work in HR department. b. List all unique department names within the company. c. List the project IDs and names of projects with a budget greater than 500,000. d. Find the names of employees along with the project IDs they are working on . e. Find the project names and budgets that employee 'E101' is working on.

Q.no 24	In an ER model, distinguish a strong entity from a weak entity and give a suitable database example.
Q.no 25	Analyze the requirements of a College Management System and identify the entities, attributes and relationships whose missing/incorrect modelling would lead to redundancy or anomalies.
Q.no 26	Compare and analyze physical and logical data independence with respect to changes made at the internal schema level and the conceptual schema level respectively, using suitable examples.
Q.no 27	Analyze why a plain File System is inadequate for a Banking application in terms of data redundancy, concurrent access, and security; justify with the specific DBMS features that resolve each problem.
Q.no 28	Given the business rules of an e-commerce Order Processing system (rules provided), analyze and identify which entities must be modelled as weak entities, justifying your choice.
Q.no 29	Critically analyze how the three-schema architecture resolves the problem of data dependence between application programs and physically stored data.
Q.no 30	Develop an EER design for a Hospital Management System containing Patient, Doctor, Department, Appointment, and Prescription. State entities, key attributes, relationships, and one suitable EER feature.
Q.no 31	Differentiate the concept of data modeling using Entity Relationship (ER) model and Enhanced ER model. Explain the additional features present in the Enhanced ER model with suitable examples.
Q.no 32	A university stores student information separately in admission, examination, and library files. Identify two problems that may occur and explain how a DBMS can address them.
Q.no 33	Design a complete ER model — using EER concepts of specialization/generalization, aggregation and multivalued attributes — for a 'Hospital Management System' covering Patients, Doctors, Departments, Treatments and Insurance; convert the model into an equivalent relational schema and justify every design decision.
Q.no 34	CASE STUDY: A university wishes to migrate its legacy file-based student record system to a DBMS. Evaluate the shortcomings of the existing system, propose a suitable three-level (three-schema) architecture for the new system, and create a complete EER diagram capturing the stated requirements, including at least one specialization hierarchy.
Q.no 35	DESIGN PROBLEM: For an 'Employee Payroll' system organized under the three-schema architecture, design sample external, conceptual and internal schema representations, and evaluate how a change in the storage structure (e.g., adding a new index) is absorbed without disturbing the conceptual or external schemas.
Q.no 36	Design an EER diagram for a 'Vehicle Rental Company' showing generalization of 'Vehicle' into 'Car', 'Bike' and 'Truck' sub-types; evaluate the trade-offs of using specialization versus a single generic entity design for this scenario.
Q.no 37	Construct an E-R diagram for a car insurance company whose customers own one or more cars each. Each car has associated with it zero to any number of recorded accidents? Also convert the E-R diagram into tables?
Q.no 38	CASE STUDY: Design a complete relational schema (with keys and referential integrity constraints) for a 'Bank' database comprising Customer, Account, Branch and Loan, based on given requirements; write the SQL DDL to create the tables and formulate three complex SQL queries (using joins, subqueries and aggregate functions) to answer specific business questions.

Q.no 39	The database consists of the following relations: • BOOK (BookID, BookName, Author, Category, Price) • MEMBER (MemberID, MemberName, Department, Phone) • ISSUE (IssueID, BookID, MemberID, IssueDate, ReturnDate) Formulate the sql query for the following: a. Write SQL CREATE TABLE statements for BOOK, MEMBER, and ISSUE, specifying suitable data types, primary keys, and foreign key constraints. b. Insert a new book into the BOOK table. c. Update the price of the book with BookID = 'B101' to ₹500. d. Delete the member record with MemberID = 'M105'. e. Display the book IDs of books issued to the member whose MemberID is 'M101'. f. Display the details of books whose price is greater than the average price of all books. g. Display books whose price is between ₹300 and ₹800. h. Write an SQL statement to add a constraint ensuring that the book price is greater than zero.
Q.no 40	CASE STUDY: A company needs a database covering Employee, Project and Department. Design the schema, express three business queries in Relational Algebra with their equivalent SQL.
Q.no 41	Two alternative SQL formulations (a correlated subquery and an equivalent JOIN) are given for the same business requirement. Evaluate which is more efficient, justify your reasoning, and additionally express the requirement using Relational Calculus.
Q.no 42	Given Student(Student_ID, Name, Program) and Enrollment(Student_ID, Course_ID), formulate equivalent relational algebra and SQL statements to retrieve names of students enrolled in Course_ID = 101.
Q.no 43	Develop SQL queries for a University database to generate an analytical report showing each department, total students, and average student marks, and include only departments with an average above 70.